The MDL Objective for Science:
Theory, Coding Scheme, and Empirical Demonstrations

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Abstract

We propose Minimum Description Length (MDL)—minimize $L(H) + L(D \mid H)$, where $L(H)$ is the description length of a hypothesis and $L(D \mid H)$ is the negative log-likelihood of the data given the hypothesis—as a formal objective function for scientific theory selection. MDL is the computable approximation to Solomonoff induction, which is Bayesian inference with the Kolmogorov/Solomonoff prior. We make four contributions. First, we argue that the history of philosophy of science from Hume through Popper, Kuhn, Lakatos, and the Bayesian tradition is best understood as a sequence of partial approximations to this objective, and identify ten cases where independent traditions converged on the same structure in different language. Second, we identify a concrete coding scheme problem: MDL requires a way of assigning description lengths to hypotheses, and we argue that the Quantified Boolean Bayesian Network (QBBN) [1] is the strongest known candidate for the logical/propositional fragment—it is complete over first-order logic, generative, tractable via belief propagation, and structurally aligned with transformer architectures, which Coppola [2] proves implement exact belief propagation on QBBN-structured factor graphs. We prove, however, that no computable coding scheme can be universally optimal: the optimal scheme exists but is uncomputable, which is the formal reason scientific inquiry is permanently open. Third, we demonstrate the framework empirically in two domains: programs and code, where a new result shows current AutoResearch systems are one $L(H)$ term short of the full MDL objective, accounting for a 65.2% improvement on benchmark problems; and logical proposition datasets, where QBBN-coded MDL selects correct conjunctive structure over disjunctive (18.4 bits saved), amortizes 8 bits of theory over 510 bits of savings at scale, and exhibits a discrete regularity crossover that instantiates Kuhn's paradigm shift mechanism at the bit level. Fourth, we argue that a system implementing the MDL objective is doing science in a formal sense regardless of substrate: a theorem-proving system with the right objective function is not merely adjacent to AGI—it is AGI.

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1 Introduction

Science has always had an objective function. It has never been written down in a single place. This paper writes it down.

The objective function is:

$$\text{minimize} \quad L(H) + L(D \mid H)$$

where H is a hypothesis, D is observed data, $L(H)$ is the description length of the hypothesis, and $L(D \mid H)$ is the description length of the data given the hypothesis. This is the Minimum Description Length (MDL) principle—the computable approximation to Solomonoff induction, which is Bayesian inference with the Kolmogorov/Solomonoff prior. It is not a new proposal. It is a formal statement of what science has always been doing informally.

This paper makes four contributions.

A synthesis of the history of philosophy of science. The history of philosophy of science from Hume through Popper, Kuhn, Lakatos, Feyerabend, and the Bayesian tradition is a single long argument about one problem: Hume's problem of induction. Every major movement is a partial approximation to the framework above. Section 6 traces this arc. Section 7 identifies ten cases where independent traditions converged on the same structure in different language.

The coding scheme problem and the QBBN as a candidate solution. MDL requires a coding scheme: a way of assigning description lengths to hypotheses. The Kolmogorov prior gives the theoretically optimal scheme but it is uncomputable. Section 4 frames the coding scheme problem precisely and argues that the Quantified Boolean Bayesian Network (QBBN) [1] is the strongest known candidate for the logical/propositional fragment of scientific hypotheses: it is complete over first-order logic (covering all of mathematics and science), generative (assigning likelihoods to proposition datasets), and tractable via belief propagation. We do not claim the QBBN solves the general coding scheme problem—cross-domain comparison (logical theories versus differential equations, for example) remains open. But for the logical fragment, the QBBN provides a concrete, computable implementation of the MDL objective. Additionally, Coppola [2] proves that sigmoid transformers implement exact belief propagation on QBBN-structured factor graphs, suggesting that the QBBN coding scheme is structurally aligned with the architecture gradient descent discovers when optimizing generative models over language.

Two empirical demonstrations. The framework is demonstrated empirically in two domains. In the domain of code and programs (Section 8), we demonstrate compression ratios, MDL model selection, the replication crisis as a missing $L(H)$ term, the search/verify asymmetry, and the Karpathy gap—a new result showing that current AutoResearch systems minimize $L(D \mid H)$ without $L(H)$, and the missing term accounts for a 65.2% improvement on benchmark problems. In the domain of logical propositions (Section 9), we use the QBBN coding scheme to show that conjunctive structure wins over disjunctive (18.4 bits), that theories are amortized over data (8 bits of theory saves 510 bits at $n = 1000$), and that MDL selects correct theory complexity at a discrete crossover that instantiates Kuhn's paradigm shift mechanism at the bit level. A single failure mode runs across both experimental domains: omitting $L(H)$ accepts overengineered code, spurious logical links, and—as argued in Section 11—worthless valid theorems. The fix is the same in every case: add $L(H)$.

Theorem proving is science; the right objective function is AGI. The critics who say automated theorem proving is not AGI because it is not science rely on an implicit theory of science that has no formal content. Under the correct theory, the verification oracle is interchangeable: a particle accelerator, a type-checker, and an MDL score calculation are all instances of the same abstract structure. A system with the right objective function is doing science regardless of substrate. Section 11 develops this argument and proposes a formal definition of AGI grounded in the framework.

Related Work

The Bayesian/Kolmogorov framework draws on Solomonoff's universal prior [15], Kolmogorov complexity [8], Rissanen's MDL principle [14], and the Bayesian tradition in philosophy of science [5]. The connection between paradigm shifts and MDL phase transitions extends Kuhn [9] and Lakatos [10]. The Bayesian brain draws on Rao and Ballard [13] and Friston [4]. The QBBN coding scheme is developed in [1]. The proof that sigmoid transformers implement exact belief propagation on QBBN-structured factor graphs is in [2]. The AutoResearch system is [7].

The `auto-research` and `mdl-science` repositories contain all code and data for the empirical demonstrations.

2 The Objective Function

2.1 The Claim

Science finds short descriptions of large bodies of observations. The right way to formalize "short description" is Kolmogorov complexity. The right way to formalize "finds" is Bayesian inference with the Solomonoff prior. Everything else in this paper is unpacking those three sentences.

The formal statement. Let H be a hypothesis—a theory, a model, a program—and let D be observed data. The objective function for science is:

$$\text{maximize} \quad \log P(H \mid D) = \log P(D \mid H) + \log P(H) + \text{const} \tag{1}$$

by Bayes' rule, where $P(D \mid H)$ is the likelihood and $P(H)$ is the prior. This is uncontroversial as a framework. The only question is: what prior?

2.2 The Prior Problem

If the prior $P(H)$ can be chosen arbitrarily, Bayes' rule is a framework, not a complete theory. Two scientists with different priors will reach different posteriors from the same data. The standard objection to Bayesianism is that this makes science subjective.

The Kolmogorov/Solomonoff answer: the prior is not arbitrary. There is a unique, principled choice grounded in the theory of computation.

2.3 The Solomonoff Prior

The Kolmogorov complexity $\mathrm{K}(x)$ of a string x is the length of the shortest program (on a fixed universal Turing machine U) that outputs x and halts:

$$\mathrm{K}(x) = \min\{|p| : U(p) = x\}.$$

Solomonoff [15] proposed assigning to each hypothesis H the prior:

$$P(H) \propto 2^{-\mathrm{K}(H)}.$$

Simpler hypotheses—those with shorter generators—receive exponentially higher prior probability. This is Occam's Razor, derived from first principles rather than assumed as a philosophical preference.

The Solomonoff prior is *universal*: for any computable probability distribution P over strings, there exists a constant c_P such that $M(x) \geq c_P \cdot P(x)$ for all strings x, where M is the Solomonoff measure. No computable prior assigns more probability to any hypothesis than M does, up to a fixed factor. The prior dominates every computable alternative.

2.4 Minimum Description Length

Because K is uncomputable (Section 3), the practical version is the Minimum Description Length (MDL) principle [14]. Instead of the shortest program, use the shortest description under a fixed code. The objective becomes:

$$\text{minimize} \quad L(H) + L(D \mid H) \tag{2}$$

where $L(H)$ is the description length of the hypothesis and $L(D \mid H)$ is the description length of the data given the hypothesis (the negative log-likelihood under a specific coding scheme).

The two terms trade off directly. $L(H)$ penalizes complexity: a theory that is too elaborate spends too many bits describing itself. $L(D \mid H)$ penalizes poor fit: a theory that predicts the data badly leaves many residual bits to transmit. The optimum is the theory that compresses the combined description most. Science is the search for short two-part codes for the world.

2.5 Occam's Razor, Derived

A consequence worth stating explicitly: two theories that predict the data equally well are not equally good. The simpler one has higher posterior. A more complex theory must predict the data *better* to compensate for its higher complexity penalty. The scientist who adds epicycles to save a theory is doing something formally wrong: increasing $\mathrm{K}(H)$ without a commensurate gain in $\log P(D \mid H)$.

The objective function is *objective* in both senses of the word: it is a target (something to minimize) and it is observer-independent (grounded in the structure of computation, not in any particular scientist's preferences). The Solomonoff prior is invariant across observers up to a fixed constant determined by the choice of universal Turing machine—not by whoever is doing the science.

3 Kolmogorov Complexity

3.1 The Intuition

Consider two strings of a thousand bits. String A alternates 0 and 1 indefinitely; String B is a typical output of a fair coin. Both are a thousand bits long. String A can be described in a few dozen characters: "alternating 0 and 1, one thousand times." String B, if genuinely random, cannot be described more briefly than writing it out in full.

Kolmogorov complexity makes this precise. The complexity $\mathrm{K}(x)$ of a string x is the length of the shortest computer program that outputs x and halts. String A has very low K; String B has K close to its own length.

Three examples locate the concept concretely. The digits of π look random but have very low K: any standard algorithm for computing π provides a program of a few hundred bytes that generates arbitrarily many digits. Newton's law of universal gravitation $F = Gm_1m_2/r^2$ is a program of a few dozen bytes that predicts the trajectories of every planet, moon, and comet we have observed, to many decimal places. A random genome, by contrast, would have K close to its raw length—but the actual human genome is substantially more compressible, because evolution is a compressive process.

3.2 The Formal Definition

Fix a universal Turing machine U. The Kolmogorov complexity of a finite binary string x with respect to U is:

$$\mathrm{K}_U(x) = \min\{|p| : U(p) = x\}$$

where $|p|$ is the length of program p in bits. In practice, the prefix-free variant (where no valid program is a prefix of another) is standard; it has cleaner information-theoretic properties.

The key theorems:

Theorem 3.1 (Invariance [8])**.** *For any two universal Turing machines U and V:*

$$|\mathrm{K}_U(x) - \mathrm{K}_V(x)| \leq c_{U,V}$$

where $c_{U,V}$ depends only on the machines, not on x. For large strings, this constant is negligible relative to $\mathrm{K}(x)$*.*

The Invariance Theorem is what makes $\mathrm{K}(x)$ objective: the choice of machine changes the value by at most a fixed overhead, independent of the string being measured. For any two computable descriptions, the complexity is the same up to a translator constant.

Proposition 3.2. *For any length n, at most 2^{n-c} strings of length n have* $\mathrm{K}(x) < n - c$*. Most strings are incompressible.*

This is a counting argument: there are only 2^{n-c} programs shorter than $n-c$ bits. When we encounter a compressible string in nature—a genome, a physical trajectory, a spectrum—that compressibility is evidence of an underlying generative process. Structure is rare.

3.3 Uncomputability: A Feature, Not a Bug

$\mathrm{K}(x)$ is not computable. No program, given x as input, reliably outputs $\mathrm{K}(x)$ and halts. The proof is a formalization of Berry's paradox: supposing K were computable, one could construct a short program that outputs a string with high complexity—a contradiction.

This is sometimes presented as a defect. It is not. Kolmogorov complexity is the *right* measure of structural complexity. Its uncomputability is the signature of universality: K captures structure defined by all possible programs, not any particular one. If K were computable, it would be reducible to some specific algorithm, and that algorithm would make implicit choices about what structure counts.

The scientist's relationship to K is the same as the mathematician's relationship to real numbers: most are uncomputable, yet the reals are the right framework. Scientists approximate K from above. Every proposed theory is an upper bound on $\mathrm{K}(D)$. The search for shorter descriptions is the search for a tighter bound. The fact that the true minimum is uncomputable is precisely why the search never ends.

3.4 MDL as the Computable Approximation

The Minimum Description Length principle [14] replaces K with a description length under a fixed coding scheme. For a parametric model H with parameters θ:

$$L(H) + L(D \mid H) = L(\theta) - \log P(D \mid \theta)$$

This is computable for any fixed model class and coding scheme. MDL is model-selection consistent: as data grows, it converges to the true model when the true model is in the hypothesis class. It reduces overfitting by penalizing complexity directly, without ad hoc regularization.

The relationship to Bayes is exact. Under a prior $P(H) \propto 2^{-L(H)}$:

$$\log P(H \mid D) = -L(H) + \log P(D \mid H) + \text{const} = -\mathrm{MDL}(H, D) + \text{const}$$

Maximizing the Bayesian posterior is equivalent to minimizing MDL, for the right choice of prior and coding scheme.

3.5 Connections to Information Theory

The expected Kolmogorov complexity of a draw from a computable distribution P equals its Shannon entropy $H(P)$ up to lower-order terms:

$$\mathbb{E}[K(X)] \approx H(X).$$

Shannon entropy is Kolmogorov complexity averaged over a distribution. Kolmogorov complexity is the pointwise version: it applies to individual strings, without requiring a distribution. When you have a single string and want to know how much structure it has—without committing to a probability model—Kolmogorov complexity is the right tool.

Lossless compression (gzip, bzip2, zstd) provides empirical upper bounds on K. A string that compresses well has low K; one that does not compress has high K. This means compression ratio is a practical proxy for structural complexity, usable without specifying any explicit model.

4 The Coding Scheme Problem

4.1 MDL Requires a Coding Scheme

The MDL objective is:

$$\text{minimize} \quad L(H) + L(D \mid H)$$

This formula is only well-defined once you specify how hypotheses and data are encoded—what "description length" means. Different coding schemes give different numerical values of $L(H)$ and $L(D \mid H)$, and in principle different rankings of theories.

The Kolmogorov framework provides the theoretically correct answer: the description length of a hypothesis is $\mathrm{K}(H)$, the length of the shortest program that generates it. The Solomonoff prior $P(H) \propto 2^{-\mathrm{K}(H)}$ is the resulting prior. This coding scheme is universal: it dominates every computable distribution up to a multiplicative constant (Section 3). For large enough data, any two universal coding schemes give the same ranking of hypotheses—the constant washes out.

The problem is that $\mathrm{K}(H)$ is uncomputable. You cannot implement the Kolmogorov coding scheme directly. Every practical MDL application uses a restricted coding scheme: a parametric family, a grammar, a neural architecture. These are computable but not universal—they cover a restricted hypothesis class and give no principled way to compare hypotheses from different families.

This is the coding scheme problem: to have a complete formal theory of science, we need a single coding scheme that is simultaneously *universal* (covering all scientific hypotheses), *generative* (assigning likelihoods to data), and *tractable* (computable in practice). No known scheme achieves all three. This section argues that the QBBN is the strongest known candidate for the logical/propositional fragment of scientific hypotheses—not a solved universal coding scheme, but a well-motivated approximation that covers a broad and important domain.

4.2 What a Practical Coding Scheme Requires

A coding scheme for scientific hypotheses must satisfy four properties to serve as a useful approximation to the Kolmogorov prior:

1. **Coverage.** The hypothesis space should cover a broad class of scientifically expressible claims. Since all of mathematics and science can be expressed in first-order logic (FOL) [1], a coding scheme that can represent any FOL formula covers the logical fragment of science—which is most of what formal science requires.

2. **Generativity.** The coding scheme must define a joint probability distribution over observations—it must assign $P(D \mid H)$ for any dataset D and hypothesis H. Without generativity, you cannot compute $L(D \mid H)$ and the MDL objective is not defined. This rules out pure theorem provers, which give truth values but not probabilities.

3. **Tractability.** The likelihoods $P(D \mid H)$ must be computable in reasonable time. Exact inference in general Bayesian networks is #P-complete; a practical coding scheme needs a tractable approximation that is empirically reliable.

4. **Structural transparency.** The coding scheme should make the structure of a hypothesis explicit—which parts are the hard logical constraints and which parts are the

learned statistical weights. This is what allows $L(H)$ to be decomposed into a structure cost and a weight cost, connecting the coding scheme to the Kolmogorov prior's notion of program length.

No existing system satisfies all four simultaneously for the full scope of scientific hypotheses. Pure FOL theorem provers satisfy (1) and (4) but not (2) or (3). Markov Logic Networks [1] satisfy (1) and (2) but not (3): inference is #P-complete. Neural networks satisfy (2) and (3) but not (1) or (4): they approximate logical structure without representing it explicitly.

4.3 The QBBN as a Candidate Coding Scheme for Logical Hypotheses

The Quantified Boolean Bayesian Network (QBBN) [1] is the strongest known candidate for a coding scheme covering logical/propositional hypotheses. It satisfies all four properties within that domain.

Coverage (FOL completeness). The QBBN paper proves that any first-order logical formula can be expressed as a QBBN factor graph. The QBBN covers the logical fragment of science—any claim that can be stated in FOL can be stated as a QBBN.

Generativity. The QBBN defines a joint probability distribution over boolean propositions via its factor graph. AND gates are deterministic (exact boolean conjunction); OR gates are learned linear exponential models that assign conditional probabilities. Together they define $P(D \mid H)$ for any dataset D of boolean observations.

Tractability. Inference in the QBBN uses belief propagation (BP) on the AND/OR bipartite factor graph. Exact BP is polynomial on trees; on loopy graphs it is an approximation. The QBBN paper reports empirical convergence in all tested cases. The bipartite AND/OR structure—alternating conjunction and disjunction layers—is the key architectural choice: conjunction nodes are deterministic, and disjunction nodes admit $O(n)$ updates under the Noisy-OR approximation.

Structural transparency. The AND/OR decomposition makes the structure of a hypothesis explicit. AND gates represent hard logical constraints—the "program" part. OR gates represent learned statistical weights—the "parameter" part. The description length decomposes as:

$$L(H) = L(\text{AND structure}) + L(\text{OR weights})$$

Each implication link costs a fixed number of bits to specify (its type and participating propositions). Each OR weight costs w bits at w-bit precision. This maps directly onto the Kolmogorov prior: AND structure corresponds to program structure (fixed overhead per instruction), OR weights to program parameters (variable, proportional to precision).

The `mdl-science` experiments in Section 9 instantiate this coding scheme directly and show that MDL in QBBN coding selects correct structure, penalizes spurious complexity by exactly its description length, and amortizes theory cost over growing datasets.

4.4 The Transformer as Independent Alignment Evidence

The coding scheme argument gains additional force from an independent direction. Coppola [2] proves that sigmoid transformers implement exact belief propagation on QBBN-structured factor graphs. Specifically, for the sigmoid transformer architecture:

- Attention heads implement AND-type joint beliefs over query-key pairs—the AND gate computation of the QBBN.
- Feed-forward networks implement OR-type disjunctive updates—the OR gate computation of the QBBN.
- The residual stream is the shared message-passing bus—the variable nodes of the QBBN factor graph.

For other transformer variants, the correspondence is approximate rather than exact, but the AND/OR bipartite structure is preserved.

This result does not prove that QBBN is the correct coding scheme for all transformers. What it suggests is that the QBBN coding scheme is not an arbitrary choice: it is structurally aligned with the message-passing architecture that gradient descent discovers when optimizing generative models over language. A transformer trained by gradient descent on next-token prediction is implicitly minimizing $L(D \mid H)$ where H has QBBN structure. The QBBN coding scheme emerges from the optimization, not from the design.

This has a direct implication for the Karpathy gap (Section 8). The gap demonstrated on toy sort programs applies, in principle, to transformer training: a model trained to minimize validation loss without an explicit $L(H)$ penalty will accumulate complexity in the OR weights whenever doing so does not hurt validation loss. Adding an explicit $L(H)$ penalty—the compressed size of the weight matrix, added to the training objective—would complete the MDL objective in QBBN coding.

4.5 The Open Problem

The QBBN is a strong candidate coding scheme for logical/propositional hypotheses. It is not a solution to the general coding scheme problem.

The open problem is cross-domain comparison. The QBBN gives a principled MDL ranking of QBBN theories against each other. It does not give a principled MDL ranking of a QBBN theory against a differential equation or a statistical regression model—because there is no shared coding scheme that covers all three. Such a ranking exists in principle: the Kolmogorov prior assigns description lengths to all computable hypotheses, and the ranking is well-defined up to a constant. But computing it requires $\mathrm{K}(H)$, which is uncomputable.

This is the computable residue of Hume's problem (Section 7). The coding scheme problem is the same impossibility in engineering form: you cannot build a universal computable coding scheme for all scientific hypotheses for the same reason you cannot solve the halting problem.

What this paper does is: identify the QBBN as the best known candidate within the logical/propositional domain; show that this domain is large (FOL covers all of mathematics and science); demonstrate empirically that MDL in QBBN coding behaves as the theory predicts; and name the open problem honestly. A universal computable coding scheme covering all scientific domains does not yet exist. The Kolmogorov framework tells us such a scheme would exist if K were computable. It also tells us K is not computable. The QBBN is the closest computable approximation currently known for the domain where science and logic intersect.

5 No Universal Computable Coding Scheme

Section 4 argued that the QBBN is the strongest known candidate coding scheme for the logical fragment of scientific hypotheses, while stating plainly that it does not solve the general coding scheme problem. This section explains why no scheme can: the optimal coding scheme exists but is not computable. The result is not a limitation of the QBBN in particular. It is a theorem about every computable scheme, and it is the formal reason science is permanently open.

5.1 The Setup

Recall the structure from Section **??**. A coding scheme assigns a description length to each hypothesis, and the MDL objective minimizes $L(H) + L(D \mid H)$ under that scheme. The Kolmogorov complexity K furnishes the ideal scheme: the Solomonoff prior $P(H) \propto 2^{-\mathrm{K}(H)}$ is *universal*, dominating every computable distribution up to a multiplicative constant. Equivalently, K lower-bounds the description length of every computable code up to a fixed additive constant: for any computable code L' there is a constant $c_{L'}$ with $\mathrm{K}(H) \leq L'(H) + c_{L'}$ for all H. The catch, established in Section 3, is that K is uncomputable.

The natural hope is that some *computable* scheme might recover universality—that we could fix a concrete, implementable code that is never worse than any competitor by more than a constant, and so deploy it as the one coding scheme for all of science. This section shows the hope is unsatisfiable.

Definition 5.1 (Computable coding scheme)**.** *A* computable coding scheme *is a total computable function $L : \mathcal{H} \to \mathbb{N}$ assigning to each hypothesis a description length, satisfying the Kraft inequality $\sum_{H \in \mathcal{H}} 2^{-L(H)} \leq 1$ (so that L is the length function of an actual prefix code).*

Definition 5.2 (Universality)**.** *A coding scheme L^* is* universal *if for every computable coding scheme L' there is a constant $c_{L'}$ such that*

$$L^*(H) \leq L'(H) + c_{L'} \quad \text{for all } H \in \mathcal{H}.$$

The constant may depend on L' but not on H. This is exactly the domination property the Kolmogorov scheme enjoys: it is no worse than any competitor, up to a fixed overhead.

We rely on two standard results from algorithmic information theory.

Theorem 5.3 (Universal schemes agree up to a constant; Li and Vitányi [**?**])**.** *Any two universal schemes differ by at most an additive constant. In particular,* K *is universal in the sense above, so any universal coding scheme L^* satisfies $|L^*(H) - \mathrm{K}(H)| \leq c$ for some constant c and all H.*

Theorem 5.4 (No computable approximation of K; Li and Vitányi [**?**])**.** *There is no computable function f and constant c such that $|f(x) - \mathrm{K}(x)| \leq c$ for all x. Kolmogorov complexity cannot be approximated to within any additive constant by a computable function.*

Theorem 5.4 is the additive-constant strengthening of the uncomputability of K proved in Section 3; it follows from the same Berry/Chaitin argument. If K could be pinned to within c bits, one could mechanically exhibit strings of certifiably high complexity and reconstruct Berry's paradox.

5.2 The Theorem

Theorem 5.5 (No universal computable coding scheme)**.** *No computable coding scheme is universal.*

Proof. Suppose, for contradiction, that L^* is a computable coding scheme that is also universal.

By Theorem 5.3, any universal scheme lies within an additive constant of K: there is a constant c with

$$|L^*(H) - \mathrm{K}(H)| \le c \quad \text{for all } H.$$

But L^* is computable by assumption. We have therefore exhibited a computable function within an additive constant of K, contradicting Theorem 5.4. Hence no computable coding scheme is universal. □

5.3 Why the Impossibility Is Generative

Theorem 5.5 closes off a particular ambition: there is no master code one could fix in advance that is guaranteed never to be beaten. The universal coding schemes that we know to exist—the Solomonoff measure and K itself—are precisely the ones that cannot be run. Universality and computability are mutually exclusive.

This is the same impossibility as Hume's problem of induction, in engineering form (Section 7). It is not a defect to be patched but the structural fact that makes the activity of science unending. If a universal computable coding scheme existed, theory selection would collapse into a finite computation: feed in the data, read off the optimal hypothesis. There would be no further discoveries to make, only a calculation to run. The theorem says no such calculation exists. Every computable coding scheme is improvable; every theory is a finite upper bound on an uncomputable optimum; the search for shorter descriptions has no terminus.

This also fixes the precise sense in which the QBBN claim of Section 4 should be read. The QBBN is not, and provably could not be, the universally optimal coding scheme—nothing computable is. What it can be, and what we argue it is, is the strongest known computable scheme for a large and well-delineated domain: the logical/propositional fragment of scientific hypotheses. Local optimality within a fixed hypothesis class, under stated desiderata, remains entirely available. It is only the global, computable, once-and-for-all code that the theorem forbids. The honest target for any coding scheme is to be the best available approximation over the domain it covers—and to be replaced, eventually, by a better one. That replacement is what scientific progress is.

6 The History of Philosophy of Science as a Single Argument

The history of philosophy of science is a single long argument about one problem: Hume's problem of induction. Every major movement is either a response to it, a refinement of a previous response, or an attempt to dissolve it. The Bayesian/Kolmogorov framework is where the argument arrives. We trace the through-line here, briefly; each movement clarifies what the framework is solving.

6.1 Hume: The Impossibility

David Hume [6] identified the deepest problem in the philosophy of science. The problem of induction: no finite set of observations logically justifies a universal law. The inference from "all observed Fs are Gs" to "all Fs are Gs" is not deductively valid. You can observe a million white swans; the next one might be black. It was, in Australia.

Worse: the obvious response—induction is justified because it has worked in the past—is itself an inductive argument. The justification is circular. Hume concluded that inductive inference is a matter of psychological habit, not reason.

6.2 Popper: The Right Asymmetry, the Wrong Conclusion

Karl Popper [11] noticed that confirmation is logically weak while falsification is logically strong. No number of white swans proves the law; one black swan refutes it. He proposed falsifiability as the demarcation criterion for science and characterized scientific progress as conjecture and refutation: bold hypotheses, then honest attempts to falsify them.

Popper was right about the asymmetry and right that unfalsifiable theories are scientifically empty. He was wrong to conclude that science involves no induction. His "corroboration" measure—how well a theory has survived tests—is, precisely defined, the Bayesian likelihood ratio $P(E \mid H)/P(E \mid \neg H)$: the update factor, stripped of the prior he refused to use. Popper computed the right quantity and declined to multiply by the prior. The Solomonoff prior completes his programme.

6.3 Kuhn: The Right Phenomenology, No Normative Theory

Thomas Kuhn [9] argued that Popper's account does not match how science actually works. Scientists operate within *paradigms*—shared frameworks of assumptions and methods—and do not abandon them at the first anomaly. When anomalies accumulate to the point where confidence breaks down, a revolution occurs: one paradigm replaces another, discontinuously.

Kuhn correctly described the phenomenology. He lacked a normative account: when *should* a paradigm shift occur? The MDL framework supplies the answer (Section 7).

6.4 Lakatos: The Right Criterion, Incompletely Formalized

Imre Lakatos [10] synthesized Popper and Kuhn via the methodology of scientific research programmes. Each programme has a *hard core* of protected assumptions and a *protective belt* of auxiliary hypotheses that absorb anomalies. A programme is *progressive* if its protective belt modifications generate new confirmed predictions; *degenerating* if they only explain known anomalies without predicting new ones.

Under MDL, a progressive programme is one where successive modifications decrease $L(H) + L(D \mid H)$ on new data. A degenerating programme is one where modifications increase $L(H)$ without decreasing $L(D \mid H)$ on new observations. Lakatos gave the right criterion; MDL makes it computable.

6.5 Feyerabend: Methodological Anarchism as the Solomonoff Prior

Paul Feyerabend [3] argued that no hypothesis should be ruled out *a priori*: science has always advanced by entertaining ideas that violated current methodological rules. "Anything goes" as a prior policy.

The Solomonoff prior assigns nonzero probability to every computable hypothesis. Nothing is excluded before the data speaks. Feyerabend's permissiveness is not anarchism—it is the correct prior policy. A prior that excludes any computable hypothesis is provably suboptimal. What Feyerabend called anarchism, Solomonoff called universality.

6.6 Bayesian Confirmation: The Culmination

Bayesian confirmation theory [5] provides the most complete response to Hume. Evidence E confirms H if and only if $P(H \mid E) > P(H)$; the degree of confirmation is the likelihood ratio $P(E \mid H)/P(E)$. This dissolves the raven paradox (observing a red fire engine confirms "all ravens are black," but only infinitesimally), handles old evidence via counterfactual reasoning, and validates Lakatos's distinction as a distinction between increasing and decreasing marginal likelihood.

Carnap spent his career trying to construct an inductive logic—a formal system that would determine the degree of confirmation of any hypothesis by any evidence. He never succeeded because he had no principled account of the prior. The Solomonoff prior completes the programme he started.

7 Ten Identifications

The directories of the `science` repo were built separately. This section is about what happens when those threads cross. The cases below are not analogies—loose, suggestive comparisons. They are *identifications*: cases where two ideas from different traditions turn out to be the same idea, expressed in different language, developed by people who were not talking to each other. The convergence is evidence that something real is being pointed at.

7.1 Hume's Problem Is the Halting Problem

Hume showed that induction cannot be logically justified: you cannot know, from finite observations, that a universal law holds. Turing showed that you cannot decide, in general, whether a program halts.

These are the same result. A universal law is a short program—a generator for an infinite class of observations. Confirming the law means certifying that the proposed program is the shortest consistent with the data. But certifying that no shorter program exists requires checking whether every shorter program eventually fails to match the data—instances of the halting problem. Since the halting problem is undecidable, $\mathrm{K}(x)$ is uncomputable, and induction cannot be deductively justified.

The reason you cannot justify induction is the same reason $\mathrm{K}(x)$ is uncomputable. Hume's problem is not a philosophical puzzle awaiting a philosophical solution. It is an undecidability result, and it has the same status as the halting problem: provably unsolvable, not merely unsolved.

What can be done—and what the Bayesian/Kolmogorov framework does—is show that the impossibility of certainty does not prevent optimal inference. Solomonoff induction is provably optimal for prediction in any computable environment. The gap between "optimal" and "certain" is exactly the gap between computable approximations to $\mathrm{K}(x)$ and $\mathrm{K}(x)$ itself.

7.2 Paradigm Shifts Are MDL Phase Transitions

Kuhn described paradigm shifts as discontinuous and partly irrational. Under MDL, they are mathematically inevitable and perfectly rational.

A paradigm is a hypothesis class—a family of theories sharing common structure. Normal science is optimization within the class: adjusting parameters, adding auxiliary hypotheses, refining the model. Each adjustment is legitimate as long as it decreases total description length.

A paradigm shift occurs when a competing hypothesis class achieves lower $L(H)+L(D \mid H)$ than the incumbent with all its protective belt modifications. The shift is discontinuous because hypothesis classes are discrete objects: you cannot smoothly interpolate between Ptolemaic and Copernican astronomy. But it is not irrational. The new paradigm wins because it compresses the data better.

The discontinuity is a phase transition in the physics sense: a sharp change in the optimal state of the system (which paradigm minimizes description length) as a continuous parameter (accumulated data) crosses a threshold. Kuhn correctly identified the phenomenology of phase transitions in scientific knowledge. MDL provides the thermodynamics underneath.

7.3 Popper's Corroboration Is Bayesian Updating Without the Prior

Popper's corroboration measure, precisely defined [12], is:

$$C(H, E) = \frac{P(E \mid H) - P(E)}{P(E \mid H) - P(E \cdot H) + P(E)}$$

This is a normalized likelihood ratio: how much more probable E is under H than under the background probability of E. It is purely a function of likelihoods. It does not involve the prior $P(H)$.

In Bayesian terms, Popper was computing exactly half of the update:

$$P(H \mid E) \propto \underbrace{P(E \mid H)}_{\text{Popper's corroboration}} \cdot \underbrace{P(H)}_{\text{refused}}$$

He refused the prior because he found subjective probability unacceptable. He was right to find it unacceptable; he was wrong to conclude the prior must be abandoned. The Solomonoff prior $P(H) \propto 2^{-\mathrm{K}(H)}$ is objective: a property of the hypothesis itself, invariant across observers. With it, corroboration becomes posterior probability, and the refusal of induction dissolves.

7.4 The Replication Crisis Is an MDL Accounting Error

The replication crisis in psychology and social science resulted from p-hacking: testing many hypotheses and reporting only those achieving $p < 0.05$. This is an MDL accounting error.

When a researcher tests N hypotheses and reports the best, the true description length of the reported hypothesis is not $L(H_{\text{reported}})$. It is:

$$L(H_{\text{reported}}) + \log_2 N$$

The extra $\log_2 N$ bits is the cost of the search: the information required to specify which of the N tested hypotheses was selected. Omitting this cost produces apparent results that do not replicate, because the apparent compression of the data was illusory.

The Bonferroni correction—adjusting the significance threshold by N—is exactly the MDL accounting correction. It has always been the right thing to do. The replication crisis is the empirical consequence of not doing it.

The reforms that followed—preregistration, larger samples, direct replication, Bayesian model comparison—are all corrections toward honest MDL accounting. Preregistration eliminates search cost from $L(H)$. Preregistered studies have lower description length overhead, and their results generalize because the $L(H)$ term is honest.

7.5 Lakatos's Progressive vs. Degenerating Is Compression Growth vs. Decay

Lakatos's distinction between progressive and degenerating research programmes, translated into MDL:

A *progressive* programme makes protective belt modifications that decrease $L(H) + L(D \mid H)$ on new data. The modifications are short (do not bloat description length) and genuinely compress new observations.

A *degenerating* programme makes modifications that increase $L(H)$— the hypothesis grows more complex—without decreasing $L(D \mid H)$ on new observations. The programme adds complexity without adding compression.

Lakatos gave the right criterion for evaluating research programmes; MDL makes it computable. The qualitative judgment (is this modification progressive or defensive?) becomes a quantitative one (does $\Delta L(H) + \Delta L(D \mid H)$ on new data decrease or increase?).

7.6 Kant's Synthetic A Priori Is Lakatos's Hard Core

Kant argued that some commitments—space, time, causality—are preconditions of experience: they cannot be revised because they are what makes experience possible. Lakatos argued that every research programme has a hard core of commitments that are methodologically protected from falsification.

These are the same idea. The difference is that Lakatos made them revisable in principle: a research programme can be abandoned, hard core and all. This is the correct update on Kant after non-Euclidean geometry falsified his specific candidates (Euclidean space as a synthetic a priori). Lakatos is Kant, empiricized.

Under the MDL framework, the hard core is the prior commitment to a hypothesis class. It is protected not by necessity but by the cost of switching to a different class (learning a new framework, retranslating existing results). The protection is finite and can be overcome by sufficient data. Kant's absolute preconditions become Lakatos's methodological conventions, which become MDL's hypothesis-class priors.

7.7 Feyerabend's "Anything Goes" Is the Solomonoff Prior

Feyerabend argued that no methodological rule holds universally: every rule that science has ever followed has been violated productively at some point. "Anything goes" as a description of scientific practice.

The Solomonoff prior assigns $P(H) \propto 2^{-\mathrm{K}(H)} > 0$ to every computable hypothesis. No hypothesis is excluded before the data speaks. This is precisely Feyerabend's permissiveness, formalized. A prior that excludes any computable hypothesis—that declares some class of ideas unscientific by methodological fiat—is provably suboptimal. The data might support the excluded hypothesis, and a reasoner that cannot entertain it incurs irreducible regret.

What Feyerabend called anarchism is, from the Kolmogorov perspective, the universality of the prior. His insight was correct; his framing was unnecessarily provocative.

7.8 Hallucination and Perceptual Illusions Are the Same Failure Mode

In predictive coding [13], the dominant framework in theoretical neuroscience, the brain is a hierarchical generative system: higher areas send predictions down the cortical hierarchy; lower areas send prediction errors up. Perception is inference: the brain computes the posterior distribution over world-states given sensory inputs, using priors built from evolution and experience.

A perceptual illusion occurs when the prior is so strong that sensory evidence cannot correct it. The hollow mask illusion is the canonical example: a hollow face mask looks convex because the prior for convex faces is overwhelming. The brain refuses to believe the correct interpretation.

In a language model, hallucination occurs when the prior over likely continuations overwhelms the grounding signal from the context. The model generates a continuation that is statistically plausible given training distribution but inconsistent with the specific facts in the prompt.

Both are the same Bayesian failure: the prior dominates the likelihood. The hollow mask illusion and language model confabulation are not analogous phenomena. They are the same computational pathology, instantiated in different substrates.

7.9 The Bayesian Brain and the Transformer Are the Same Architecture

Predictive coding models the brain as performing belief propagation on a hierarchical factor graph [4]: predictions propagate top-down, prediction errors propagate bottom-up, and the system minimizes free energy (variational approximation to surprise).

The transformer implements exact belief propagation on a specific factor graph [2]: the attention mechanism computes AND-type joint beliefs over query-key pairs; the feed-forward network computes OR-type disjunctive updates; the residual stream is the shared message-passing bus.

These are not analogies. They are the same algorithm instantiated on different graphs. The brain and the transformer are doing the same computation. The differences are in the graph structure, the learning algorithm, and the substrate—not in the fundamental operation.

7.10 AutoResearch Is Solomonoff Induction Made Computable and Distributed

Solomonoff induction is uncomputable but is the provably optimal prediction strategy. Karpathy's AutoResearch system [7] is a computable, distributed approximation to it.

The correspondences are structural. The hypothesis space is the space of training scripts and model configurations—a computable, enumerable space. The reward signal (validation loss, benchmark accuracy) approximates $P(D \mid H)$: lower loss means higher likelihood of the training data under the modified hypothesis. The keep/discard decision approximates the Bayesian posterior update. The agent's tendency to propose local modifications first is an implicit Kolmogorov prior: smaller modifications correspond to lower description length.

The key insight Karpathy identified—finding a good result is hard, but verifying one is cheap—is exactly the search/verify asymmetry in Solomonoff induction. The search over hypothesis space is expensive; checking whether a proposed hypothesis compresses the data better is cheap. The distributed, SETI@Home-style architecture parallelizes the search over hypothesis space.

Crucially, AutoResearch accepts contributions from untrusted sources. A modification is evaluated by whether it reduces description length of the training data—a property of the modification itself. The source is irrelevant. This is the Kolmogorov principle applied to institutions: *trust the object, not the messenger.*

8 Experiments I: Code and Programs

The experiments in this section instantiate the MDL objective function in the domain of programs and code. All results are reproducible from the `auto-research` repository. Six experiments, each isolating one aspect of $L(H) + L(D \mid H)$: `compress` measures $L(H)$ directly via compression ratio; `mdl` shows the $L(H)$ vs $L(D \mid H)$ tradeoff in polynomial model selection; `replicate` shows what happens when $L(H)$ is underreported; `search` shows that verification is evaluating $L(D \mid H)$ cheaply; `update` shows the Kolmogorov prior doing work data alone cannot; `autoloop` runs the full MDL loop and identifies the gap between current AutoResearch systems and the complete objective.

8.1 Structure Is Compressibility (`ar compress`)

Kolmogorov complexity is uncomputable, but lossless compression provides a computable upper bound. A string that compresses well has low K; one that does not compress has high K. The compression ratio approximates $\mathrm{K}(x)/|x|$—the complexity relative to the raw length.

File	Raw bytes	Ratio	Interpretation
`repeated_pattern`	1009	0.036	highly structured
`genome_like`	766	0.123	highly structured
`source_code`	868	0.414	moderate structure
`english_prose`	1501	0.502	moderate structure
`pi_digits`	1009	0.520	moderate structure
`random_noise`	369	0.564	low structure
`natural_law`	591	0.604	low structure

The genome compresses to 12.3% of its raw size. DNA looks complex but is built from repeated motifs; evolution is a compressive process.

The digits of π compress to only 52.0%. π has very low true Kolmogorov complexity—there exists a short program that generates it exactly—but gzip finds only local repetitions. This illustrates the gap between the compression proxy and the true $\mathrm{K}(x)$: gzip is an upper bound, not the true complexity.

Natural laws compress to 60.4%, below random noise. The structure in natural laws is semantic, not syntactic; gzip cannot see it. The true K of Newton's law—the shortest program predicting all observations it covers—is extremely low.

8.2 The MDL Tradeoff Is Real (`ar mdl`)

MDL model selection fits polynomials of degree 1 through 8 to data from a known degree, computing $L(H) + L(D \mid H)$ for each.

On a true degree-2 parabola, 40 points, $\sigma = 0.3$:

Degree	$L(H)$	$L(D \mid H)$	Total
1	64.0	101.2	165.2
2	96.0	−0.2	**95.8**
3	128.0	−0.4	127.6
4	160.0	−0.4	159.6

MDL recovers the true degree. The decisive test is `overfit_trap`: same generator, only 12 noisy points ($\sigma = 1.2$):

Degree	$L(H)$	$L(D \mid H)$	Total
1	64.0	31.4	**95.4**
2	96.0	25.4	121.4
3	128.0	24.9	152.9

MDL chooses degree 1, not the true degree 2. This is not a failure. With 12 noisy points, the improvement in fit does not justify the 32-bit complexity penalty. The framework is being epistemically honest: we do not have enough data to establish the true degree. Science should say the same thing.

8.3 The Replication Crisis Is One Missing Term (`ar replicate`)

Simulating p-hacking on null data:

Tests N	Evidence (bits)	Search cost $\log_2 N$	Corrected	Verdict
10	5.13	3.32	1.80	not significant
50	7.28	5.64	1.63	not significant
100	6.28	6.64	−0.37	not significant

Every naive result clears the 4.32-bit threshold ($p < 0.05$). Every corrected result is noise. A single test on data with a real effect of size 0.8 produces 16.45 bits; at $N = 100$ tests the corrected value is 21.25 bits—the real effect survives with massive margin. The search cost $\log_2 N$ is exactly the Bonferroni correction in bits. The replication crisis is the empirical consequence of omitting this term from $L(H)$.

8.4 Verification Is Constant; Search Scales (`ar search`)

Candidate	Tests passed	Verify time (ms)
`broken_empty_return`	0/8	0.13
`broken_mutates_input`	0/8	0.62
`broken_off_by_one`	0/8	0.09
`correct_bubble`	8/8	0.07
`correct_recursive`	8/8	0.08

Verification time is flat. Complexity and correctness do not affect verification cost. Search cost accumulates at 6.1× the verification cost for 5 candidates; with 50,000 candidates the ratio is ∼60,000×. The verification cost never moves. The candidate `broken_returns_none` is caught in 0.07 ms with no knowledge of its source or intent. The object speaks for itself.

8.5 The Right Prior Converges Faster (`ar update`)

Four hypotheses—$3x$, $3x + 0{\cdot}x^2$, $3x + 0{\cdot}x^3$, $3x + 0{\cdot}x^2 + 0{\cdot}x^3$—predict identically for every input.

Step	Uniform on $3x$	Kolmogorov on $3x$
Prior	25.0%	99.6%
1	25.0%	99.6%
5	25.0%	99.6%

The uniform prior stays at 25% forever. The Kolmogorov prior starts at 99.6% before any data. Among hypotheses that predict equally well, the prior decides—and it should be the Kolmogorov prior. In the weak-signal case (true $2x$, noise = 1.5), the Kolmogorov prior starts $2x$ lower than uniform (17.4% vs 25%) yet reaches 78% at step 3 versus 68.5% for uniform. The mechanism: the Kolmogorov prior suppressed the wrong alternatives from the start.

8.6 The Karpathy Gap (`ar autoloop`)

The `autoloop` experiment models the Karpathy AutoResearch loop in two modes: full MDL ($L(H) + L(D \mid H)$) and $L(D \mid H)$ only. Starting from a baseline that already passes all 8 tests:

Iter	Mutation	Tests	$L(H)$	$L(D\|H)$	MDL	Decision
0	baseline	8/8	279	0.0	279.0	—
1	`01_fix_break`	8/8	257	0.0	257.0	keep
2	`02_add_type_check`	8/8	315	0.0	315.0	discard
3	`03_simplify`	8/8	170	0.0	170.0	keep
4	`04_introduce_bug`	3/8	197	15.0	212.0	discard
5	`05_use_builtin`	8/8	97	0.0	97.0	keep
6	`06_overengineered`	8/8	211	0.0	211.0	discard

Full MDL: baseline 279.0 → 97.0, **65.2% reduction**. $L(D \mid H)$-only: 0 improvements, stuck forever. The decisive case is `06_overengineered`: passes all tests, $L(D \mid H) = 0$, $L(H) = 211$. Full MDL discards it; $L(D \mid H)$-only cannot distinguish it from `05_use_builtin`.

This is the Karpathy gap: Karpathy's `val_bpb` is $L(D \mid H)$ in QBBN coding (Section 4) without $L(H)$. The gap is not a toy artifact. Coppola [2] proves that sigmoid transformers implement exact BP on QBBN-structured factor graphs, so every transformer trained to minimize validation loss without an explicit complexity penalty is running the same incomplete objective. The missing term is $L(H)$. Adding it completes MDL and prevents accumulating complexity over hundreds of iterations whenever doing so does not hurt validation loss.

9 Experiments II: Logical Propositions

The experiments in this section instantiate the MDL objective function in the domain of logical proposition datasets, using the QBBN coding scheme developed in Section 4. All results are reproducible from the `mdl-science` repository.

A dataset is a list of boolean proposition observations. A theory is a set of *implication links* (premise → conclusion, firing when the premise is true) and *conjunctive links* ($[\text{premise}_1, \ldots, \text{premise}_n] \to$ conclusion, firing only when all premises are true). These correspond directly to the QBBN: conjunctive links are AND gates, implication links are OR-gate inputs.

The two-part code is concrete. $L(H)$ counts the number of links times the bits per weight—a fixed cost per link, independent of dataset size. $L(D \mid H)$ is the total negative log-likelihood of the observed propositions under the theory's conditional probabilities, with link weights calibrated empirically from the data. Propositions not covered by any link fall back to their empirical base rate, ensuring the theory can only help—never hurt—on uncovered propositions.

What the experiments show and do not show. These experiments score a fixed set of candidate theories; they do not perform open search over the space of all possible QBBN structures. The MDL objective correctly selects among the candidates provided. The next step—enumerating candidate theories from data without a predetermined candidate set—is the automated theory discovery problem, which the framework defines but does not yet solve at scale. This is the logical-domain analogue of what AutoResearch does in the code domain: propose candidates, score them, keep improvements. The `mdl-science` experiments implement the scoring step; candidate generation is future work.

9.1 Conjunctive Structure Wins (`mdl compress`)

Data where $r = p \wedge q$ exactly ($n = 50$ triples). Six theories compete. Bits saved is positive when the theory beats the no-theory baseline, negative when it is worse:

Theory	$L(H)$	$L(D \mid H)$	MDL	Bits saved
null (no links)	0.0	143.5	143.5	0.0
one link: $p \to r$	8.0	134.2	142.2	+1.3
one link: $q \to r$	8.0	140.0	148.0	−4.5
two links: $p \to r$, $q \to r$	16.0	141.8	157.8	−14.3
conjunctive: $p \wedge q \to r$	8.0	117.1	125.1	**+18.4**
wrong link: $p \to q$	8.0	143.2	151.2	−7.7

Bits saved = MDL(null) − MDL(theory).

The conjunctive theory wins by 18.4 bits using only 8 bits to describe one link. The disjunctive two-link theory loses by 14.3 bits despite having both premises available: it misfires when only one of p or q is true, predicting $r = 1$ when $r = 0$. The wrong link loses by 7.7 bits. Even the partial theory $p \to r$ barely wins at +1.3 bits.

The key result: having more links is not always better. The two-link disjunctive theory has strictly more information about r than the one-link conjunctive theory, yet it performs worse. What matters is whether the links capture the true generative structure. The disjunctive theory has the right variables but the wrong logical connective.

This result is a direct consequence of the QBBN coding scheme. The AND gate is a zero-parameter, deterministic structure that adds only its own description to $L(H)$. When the true structure is conjunctive, the conjunctive coding is maximally efficient: it pays zero for the AND computation itself, only for the single weight that says how strongly the conjunction predicts the conclusion.

The $L(H)$ ablation. The result for T3 in the compete experiment below makes this precise: the spurious link has $L(D \mid H)$ identical to the correct theory but $L(H)$ higher by exactly 8 bits. Omitting $L(H)$ from the objective would make the two theories indistinguishable. This is the logical-domain Karpathy gap: just as omitting $L(H)$ in the code domain accepts overengineered programs, omitting $L(H)$ in the logical domain accepts spurious links. The failure mode is the same across substrates.

9.2 Theories Are Amortized Over Data (`mdl scale`)

One conjunctive link, $L(H) = 8$ bits, regularity = 1.0:

n	$L(D)$	MDL	Bits saved	Comp.%
5	13.3	18.7	−5.4	−40.2%
10	28.8	31.6	−2.8	−9.7%
25	70.9	65.8	+5.0	+7.1%
50	136.7	118.1	+18.5	+13.6%
100	276.2	231.2	+44.9	+16.3%
200	535.7	440.5	+95.2	+17.8%
500	1326.8	1079.0	+247.9	+18.7%
1000	2680.7	2170.1	+510.6	+19.0%

Bits saved = L(D)− MDL.

$L(H) = 8$ bits at every scale. The theory loses at $n = 5$ and $n = 10$, crosses into positive territory between $n = 10$ and $n = 25$, and saves 510 bits at $n = 1000$—64 times its own description length. The crossover point (theory overhead recovered at $n \approx 15$ pairs) is precisely measurable because the QBBN coding scheme makes $L(H)$ genuinely independent of dataset size.

This is the formal version of the intuition behind scientific laws. Newton's laws cost a fixed number of bits to state; every new observation is compressed for free once the theory exists. The amortization is not a metaphor. It is a computable quantity, demonstrated here at the level of individual bits.

9.3 MDL Selects Correct Theory Complexity (`mdl compete`)

Four theories at three regularity levels. T0 = null, T1 = one implication link, T2 = correct conjunctive link, T3 = T2 plus one spurious implication link:

Theory	$L(H)$	$L(D \mid H)$	MDL	Bits saved
regularity = 1.0				
T0: null	0.0	276.2	276.2	0.0
T1: one link	8.0	272.6	280.6	−4.5
T2: correct conjunctive	8.0	223.2	231.2	**+44.9**
T3: T2 + spurious link	16.0	223.2	239.2	+36.9
regularity = 0.5				
T0: null	0.0	272.5	272.5	**0.0**
T1: one link	8.0	272.0	280.0	−7.6
T2: correct conjunctive	8.0	262.6	270.6	+1.9
T3: T2 + spurious link	16.0	262.6	278.6	−6.1
regularity = 0.0				
T0: null	0.0	273.5	273.5	**0.0**
T1: one link	8.0	273.5	281.5	−8.0
T2: correct conjunctive	8.0	273.5	281.5	−8.0
T3: T2 + spurious link	16.0	273.5	289.5	−16.0

Bits saved = MDL(T0) − MDL(theory).

At regularity = 1.0, T2 saves 44.9 bits. T3 saves 36.9 bits— exactly 8 bits less. The spurious link costs exactly its description length and changes $L(D \mid H)$ not at all. This is Occam's razor in its precise form: *a theory that is correct but unnecessarily complex loses to the minimal correct theory by exactly the description length of the excess.*

At regularity = 0.5, T2 barely saves 1.9 bits; T3 cannot recover the extra 8-bit cost. At regularity = 0.0, T0 wins; T1 and T2 tie at exactly -8.0 bits; T3 pays -16.0 bits. The framework penalizes each spurious link by exactly its description length regardless of whether the link actively harms prediction.

The triad. The T2/T3 result here joins a pattern that runs across all three experimental layers of this paper. Omitting $L(H)$ from the MDL objective produces the same failure mode in every domain: in the code domain (Section 8), it accepts the overengineered sort implementation that passes all tests but costs 114 bits more than the minimal correct solution; in the logical domain here, it accepts the spurious link that compresses no better but costs 8 bits more; and in the theorem-proving domain (Section 11), it would accept any valid theorem regardless of whether it compresses the mathematical corpus. The failure mode is substrate-independent. So is the fix: add $L(H)$.

The regularity crossover is the MDL phase transition of Section 7 instantiated at the bit level. As regularity increases from 0 to 1, the optimal theory switches discontinuously from T0 to T2. T2 goes from -8.0 bits (noise) to $+1.9$ bits (weak signal) to $+44.9$ bits (perfect regularity). The optimal hypothesis class jumps when the theory's amortized savings first exceed its description length cost. This is exactly what Kuhn observed in the history of science and what Lakatos formalized as the distinction between progressive and degenerating research programmes. Here it is a measurable threshold in bits.

10 Verification and the Present Moment

10.1 The Search/Verify Asymmetry

The objective function tells us what science is optimizing. Running the optimization loop requires solving a different problem: given a proposed hypothesis H, how do you verify that it actually advances the objective?

The key structural fact:

> Finding a good H is hard. Verifying a proposed H is cheap.

This asymmetry is not incidental. It is a feature of the Kolmogorov framework, and it is the design principle behind every sound verification protocol. Finding the shortest program for x requires searching an infinite space. Checking whether a proposed program outputs x requires only running it.

10.2 Algorithmic Verification

For claims in the verifiable domain, verification can be automated.

Formal proof checking. If a hypothesis is expressed as a theorem with a machine-checkable proof (Lean 4, Coq, Isabelle), verification is linear in proof length. The checker evaluates the *object*, not the source. A proof submitted by an unknown agent is as valid as one submitted by a Fields Medalist, if it type-checks. This is the purest instantiation of the Kolmogorov principle.

Empirical reproduction. Given code, weights, and data, rerun the experiment. Does the reported metric reproduce? This is cheap relative to the original research cost and provides a binary signal.

Compression-based verification. Does the proposed theory compress the data better than the incumbent? MDL makes this a computable number. Given a fixed coding scheme, compute $L(H) + L(D \mid H)$ for the new theory and for the incumbent. If the new value is lower, the theory advances the objective. No subjective judgment required.

10.3 The Untrusted Source Principle

The most important design principle for any verification system:

> The verifier does not need to trust the source.

Under the Kolmogorov framework, a hypothesis is evaluated on its length and its predictive accuracy—properties of the object itself. The identity, credentials, or intentions of whoever proposed it are irrelevant to its truth.

This is an engineering requirement, not just a philosophical point. A distributed AutoResearch system cannot verify the identity or intent of every contributing agent. But if verification is a property of the object (does the proof type-check? does the loss decrease? does the description length shrink?), the system can accept submissions from arbitrary untrusted sources and remain sound. The math is the trust anchor.

10.4 Why This Matters Now

Two developments make the framework more than a philosophical exercise.

Autonomous AI research. Systems like Karpathy's AutoResearch close the optimization loop at the level of ML experiments: propose a modification, verify whether it reduces the loss, keep it if it does, repeat. This is Solomonoff induction approximated by the only means available: a computable, distributed, heuristic search over hypothesis space, with cheap verification at each step.

Machine-verified formal proof. Large proofs in Lean 4 and Coq are being produced and verified at scale. The verification is automated; the search (finding the proof) is expensive; the check (type-checking) is cheap. The asymmetry is the same.

Both represent the first time the optimization loop of science can be partially automated. The objective function—minimize $L(H) + L(D \mid H)$— is the same as it has always been. What is new is the infrastructure to approximate it at scale.

10.5 Conclusion

The history of philosophy of science is a record of humanity circling a single structure—the problem of optimal inference in a computable world—without the mathematical tools to see it directly. Hume identified the core impossibility. Popper, Kuhn, and Lakatos each described a facet of the solution without being able to state it. The Bayesian tradition formalized the update rule. Kolmogorov and Solomonoff formalized the prior.

The tools now exist to state the full solution. Science is the distributed search for hypotheses that minimize description length. Verification is cheap relative to discovery. The prior is the Solomonoff prior. The update rule is Bayes' rule. The objective function is MDL.

Everything else is details.

11 Automated Theorem Proving Is Science

11.1 The Critics' Claim

A recurring objection in discussions of artificial general intelligence runs as follows. Automated theorem proving—even at superhuman scale, even across all of mathematics—is not AGI, because formal reasoning is not the same as doing science. Science requires engaging with the empirical world: forming hypotheses, running experiments, updating on data. A system that only manipulates symbols inside a formal system is doing something categorically different from what scientists do. Theorem proving, however impressive, is a party trick. It is not intelligence.

This objection is widespread and sounds compelling. We argue it is wrong, and that it is wrong for a precise reason: it rests on an implicit theory of science that has never been written down—and when you write it down, it does not survive scrutiny.

11.2 The Implicit Theory Being Assumed

The critics' argument requires a theory of what science is, specifically one that excludes formal reasoning. The theory being assumed, roughly, is this: science is the activity of interacting with the physical world through observation and experiment, and formal reasoning is merely a tool that science uses, not science itself.

This theory has intuitive appeal but no formal content. It cannot tell you:

- When two competing theories are equally consistent with the data, which is better?
- How much complexity is a theory allowed before it is no longer parsimonious?
- When should a paradigm be abandoned in favor of a competitor?
- What counts as a legitimate auxiliary hypothesis versus an ad hoc patch?

These are the central questions of the philosophy of science, and the informal "interaction with the physical world" theory has no answers to them. It is not a theory of science. It is a description of one setting in which science sometimes happens.

11.3 The Formal Theory

The Bayesian/Kolmogorov framework provides an actual theory of science, with formal answers to all of the above questions. Restating it precisely:

Definition 11.1 (Scientific activity)**.** *A process is engaged in scientific activity if it:*

1. *Maintains a space of computable hypotheses $\mathcal{H}$;*

2. *Assigns prior probabilities $P(H) \propto 2^{-\mathrm{K}(H)}$ to hypotheses, preferring shorter descriptions;*

3. *Evaluates hypotheses against observations via a likelihood function $P(D \mid H)$;*

4. *Updates beliefs by Bayes' rule: $P(H \mid D) \propto P(D \mid H) \cdot P(H)$; and*

5. *Selects hypotheses that minimize $L(H) + L(D \mid H)$.*

Notice what is *not* in this definition: laboratories, physical instruments, empirical data in any particular form, or any requirement that D come from the external world rather than from a formal system. The definition is about the *structure* of the inference, not the *substrate* in which it occurs.

11.4 The Verification Oracle Is Interchangeable

The key insight is that the search/verify asymmetry (Section 10) holds regardless of what the verifier is.

In empirical science, the verifier is experiment: propose a hypothesis, derive a prediction, run the experiment, check whether the prediction holds. Finding a good hypothesis is hard; running the experiment is cheap relative to the search.

In formal mathematics, the verifier is the proof checker: propose a theorem and a proof, and the type-checker verifies it in time linear in proof length. Finding a good proof is hard; checking it is cheap.

In logical proposition learning—the domain of the `mdl-science` experiments (Section 9)—the verifier is the MDL score: does the proposed theory actually compress the observed propositions? At regularity = 0.0, no theory compresses the data. At regularity = 1.0, the correct conjunctive theory wins at 44.9 bits saved. The score is cheap, deterministic, and oracle-agnostic.

All three are instances of the same abstract structure:

$$\mathrm{Search}(H \in \mathcal{H}) \xrightarrow{\text{expensive}} \hat{H} \xrightarrow{\text{cheap verification}} \text{accept or reject}$$

The oracle that implements "cheap verification" differs in its physical instantiation—a particle accelerator, a type-checker, an MDL score calculation—but the computational structure is identical. A system that can search hypothesis space and verify proposals is doing science. The nature of the oracle is irrelevant to this classification.

11.5 The $L(H)$ Triad

A single failure mode runs across all three experimental layers of this paper, and it is the same failure mode that infects the critics' implicit theory of science.

Omitting $L(H)$ from the MDL objective:

- In the **code domain** (Section 8): accepts the overengineered sort implementation that passes all tests but costs 114 bits more than the minimal correct solution. The $L(D \mid H)$-only loop cannot distinguish it from the optimal.

- In the **logical domain** (Section 9): accepts the spurious link (T3) that compresses no better than the correct theory (T2) but costs 8 bits more. The $L(D \mid H)$-only scorer cannot distinguish T2 from T3.

- In the **theorem-proving domain**: accepts any valid theorem regardless of whether it compresses the mathematical corpus. A theorem prover without $L(H)$ has no basis for preferring a proof that unifies ten results over one that merely restates what was already known.

The fix is the same in every domain: add $L(H)$. The critics who argue that theorem proving is not science are, without knowing it, describing a system that is missing $L(H)$—a system that can verify but cannot prefer, that can prove but cannot compress. Adding $L(H)$ is not a minor technical adjustment. It is the difference between a system that can do science and one that cannot.

11.6 What Formal Mathematics Looks Like Under This Theory

Under the Kolmogorov framework, a mathematical theorem is a hypothesis about the structure of a formal system. A proof is the verification. The "observations" D are the axioms and previously established theorems. The "experiment" is type-checking.

Consider what a mathematician does:

- She maintains a space of possible theorems (hypotheses).
- She assigns higher prior probability to theorems that are short, elegant, and follow from a few strong principles—precisely $P(H) \propto 2^{-\mathrm{K}(H)}$.
- She evaluates a candidate theorem against the existing body of mathematics (the "data").
- She updates: a theorem that follows from deep principles and connects many areas is more valuable than one that requires a long, ad hoc proof.
- She publishes results that compress the existing body of knowledge—unifying previously separate results, revealing that two seemingly different theorems are instances of one.

The `scale` experiment (Section 9) demonstrates the amortization argument concretely: 8 bits of theory saves 510 bits at $n = 1000$ observations. Mathematical theorems are the extreme case—a single theorem can compress an infinite class of observations into a proof of bounded length. The mathematician is doing exactly what the physicist does. The substrate differs; the objective function is the same.

11.7 The Objection Answered

The critics are right that theorem proving, as typically practiced, is not the same as doing science. But this is because automated theorem provers as currently deployed are missing $L(H)$. They are optimized to prove specified theorems, not to search for theorems worth proving. They are given H and asked to verify it, rather than searching $\mathcal{H}$ for the H that minimizes description length.

This is an engineering gap, not a conceptual impossibility. A theorem-proving system equipped with:

1. A prior over theorems weighted by description length,

2. A likelihood that measures how much a theorem compresses and unifies the existing mathematical corpus,
3. A search procedure over the space of provable statements, and
4. A cheap verifier (the type-checker),

would be doing science in the full formal sense. The `compete` experiment shows what this looks like at miniature scale: T3 loses to T2 by exactly 8 bits at every regularity level. A theorem-proving system with the MDL objective would prefer the shorter proof of the same theorem, prefer the theorem that unifies two results over the one that merely restates one, and discard conjectures that cost more to state than they save in compression. This is not a description of an exotic future system. It is a description of what good mathematicians already do.

11.8 AGI and the Objective Function

Proposition 11.2 (AGI as correct objective function)**.** *A system exhibits artificial general intelligence if and only if it implements—exactly or approximately—Solomonoff induction: maintaining a prior over computable hypotheses weighted by description length and updating by Bayes' rule on available observations.*

This definition is formal, precise, substrate-independent, and graded: systems can approximate Solomonoff induction more or less closely, with better approximations constituting more general intelligence. It connects AGI directly to the objective function for science: a system doing science in the formal sense is, by this definition, exhibiting general intelligence.

Under this definition, the question "is theorem proving AGI?" has a precise answer: it depends on whether the system has the right objective function. A system given theorems to prove and proving them is not AGI—it is a capable special-purpose tool. A system that searches for theorems worth proving, verifies them, and updates its prior—minimizing description length over the space of mathematical knowledge—is AGI, regardless of whether it ever runs a physical experiment.

11.9 The Broader Implication

The tools to build such a system exist. The objective function is written down in this paper. The verification infrastructure exists: Lean 4 type-checks proofs in linear time; benchmark evaluation checks model performance in seconds; experimental platforms check empirical predictions in hours or days. The search infrastructure is being built: AutoResearch [7], agent-based research systems, and large language models used as hypothesis generators are all approximating the search step. The QBBN coding scheme (Section 4) provides a concrete implementation of the MDL objective for the logical domain.

What has been missing is the theoretical unification: the recognition that all of these are the same process, approximating the same objective function, and that a system implementing that objective function in any verifiable domain is doing science and exhibiting general intelligence.

That recognition is the contribution of this paper.

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